Internet of Things Course (420-521-VA)

Final Project
Phase 1

Due Date:

Thursday, October 2nd, 2025

Team Members (Group 5)

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Overview

Phase 1 of the IoT Smart Store project requires:

1. A web page to add customers.
2. A database design for storing customer data.
3. Confirmation of successful inserts via a blue LED.
4. Error handling via a red LED and buzzer.
5. In-page notifications for both cases.

Our group implemented this using a Flask web application, SQLite database, and Raspberry Pi GPIO for LED/buzzer control.

System Design

Database Schema

Table: Customers

- customer_id (PK, Auto Increment)
- first_name (TEXT, NOT NULL)
- last_name (TEXT, NOT NULL)
- email (TEXT, Unique, NOT NULL)
- date_created (TIMESTAMP)

Data Flow

Adding Customer

1. User enters customer info on web form.
2. Flask backend inserts data into database.
3. On success -> Blue LED ON + UI notification.
4. On error -> Red LED + Buzzer + error message.

Deleting Customer

1. User finds and deletes desired customer in list.
2. Flask backend deletes data using associated ID.
3. On success -> Blue LED ON + UI notification.
4. On error -> Red LED + Buzzer + error message.

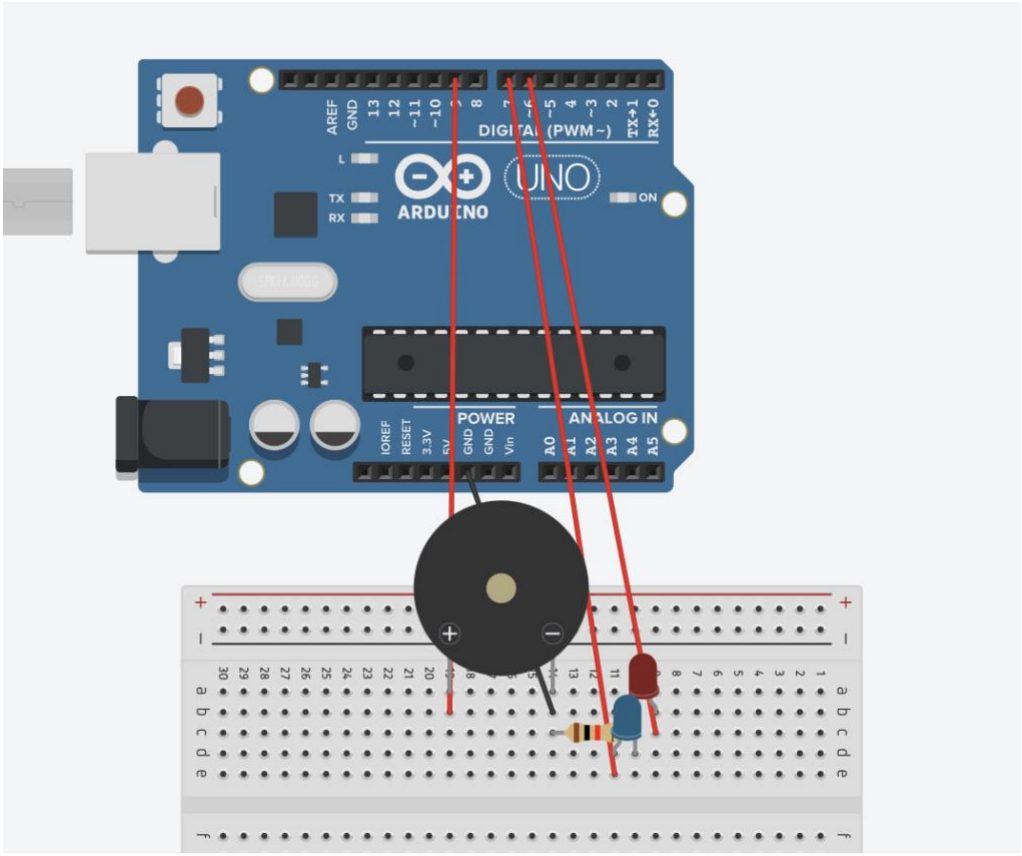
Implementation

- Framework: Flask (python)
- Database: SQLite
- Frontend: HTML/CSS form with simple UI
- Hardware: Raspberry Pi, LEDs, resistors, buzzer, breadboard, wires
- Repository: https://github.com/Crisguts/IoT-Semester-Project-

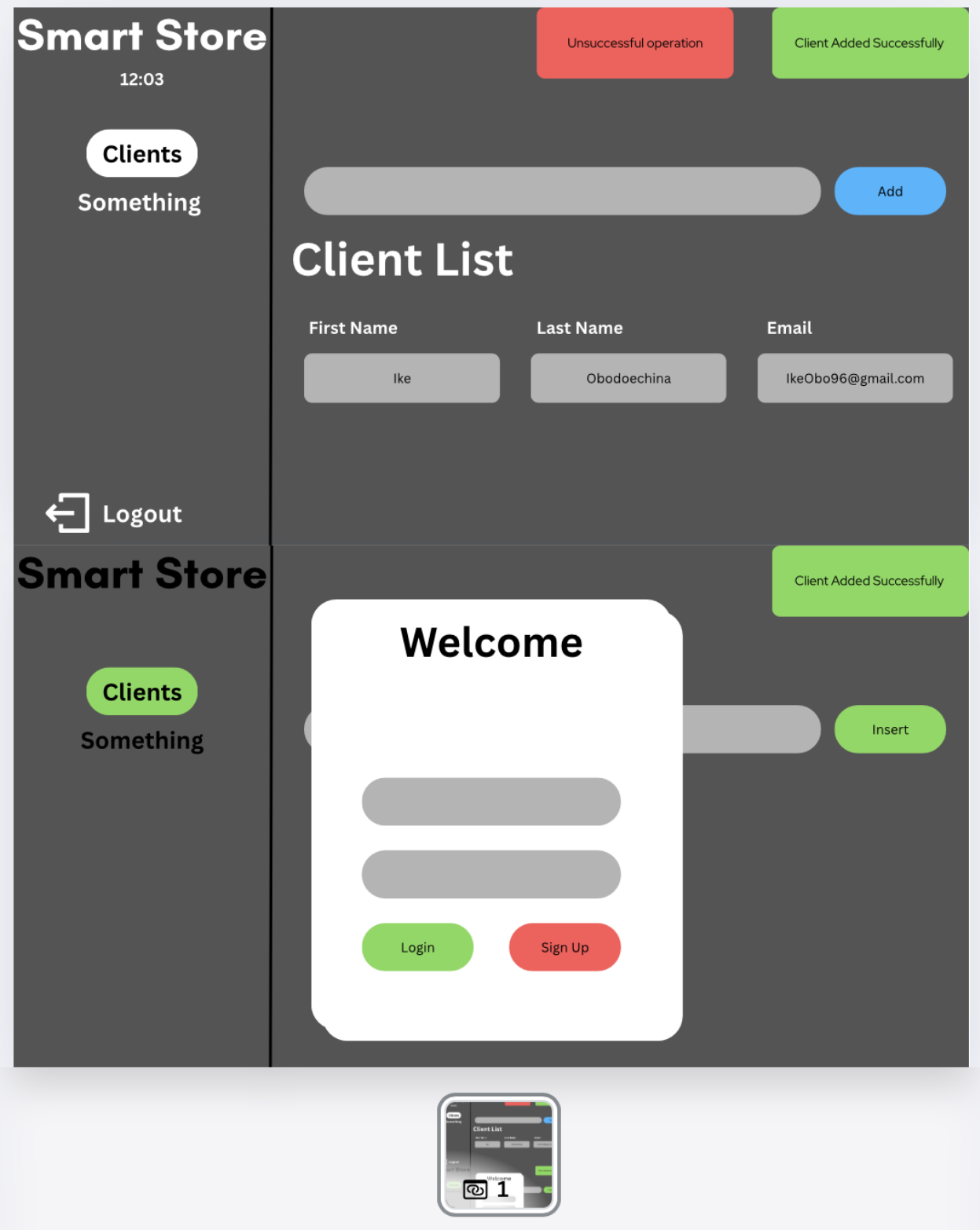
Task Distribution

Team Member	Task
Cristian	Implemented the Flask backend application, including database connection, CRUD functionality, and input validation. Developed GPIO code for LED and buzzer and completed the wiring design. Prepared project documentation, phase 1 document and README (step-by-step tutorial). Managed GitHub repository structure and commits.
Trevor	Designed the UI/UX layout using Canva and translated it into HTML/CSS templates (index.html) with Bootstrap. Integrated a JavaScript real-time clock and implemented dynamic display of customer records on the page. Ensured the interface aligned with user-friendly design principles.
Parth	Designed and tested the Customers database table. Defined schema fields, created and validated the database structure, and ensured proper integration with the Flask app. Pushed changes to the shared repository for team integration.

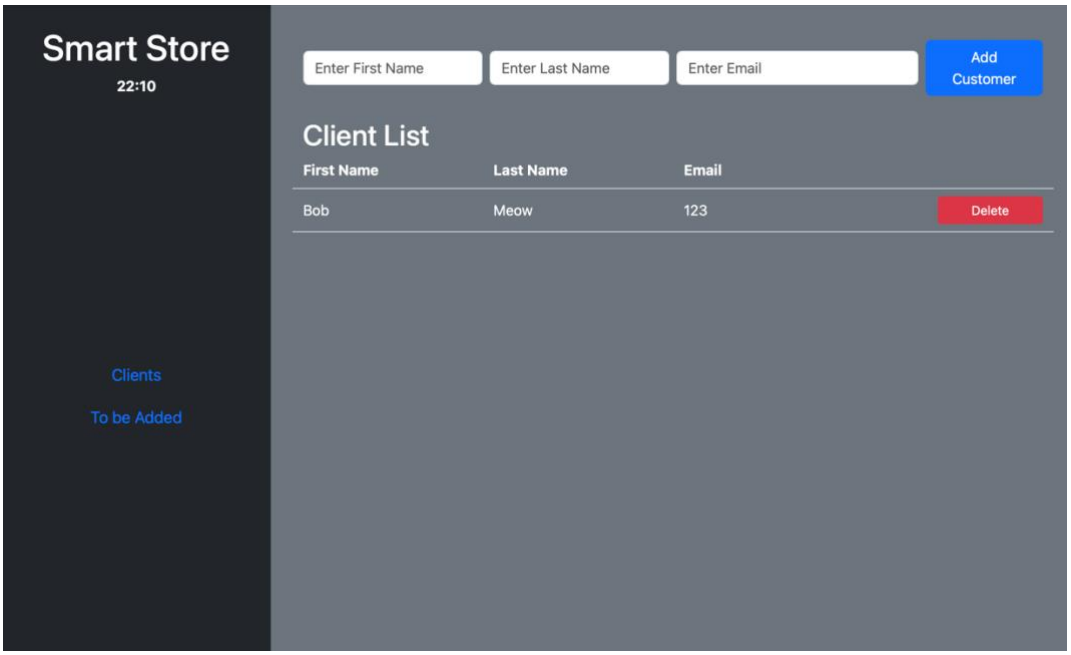
Circuit Design



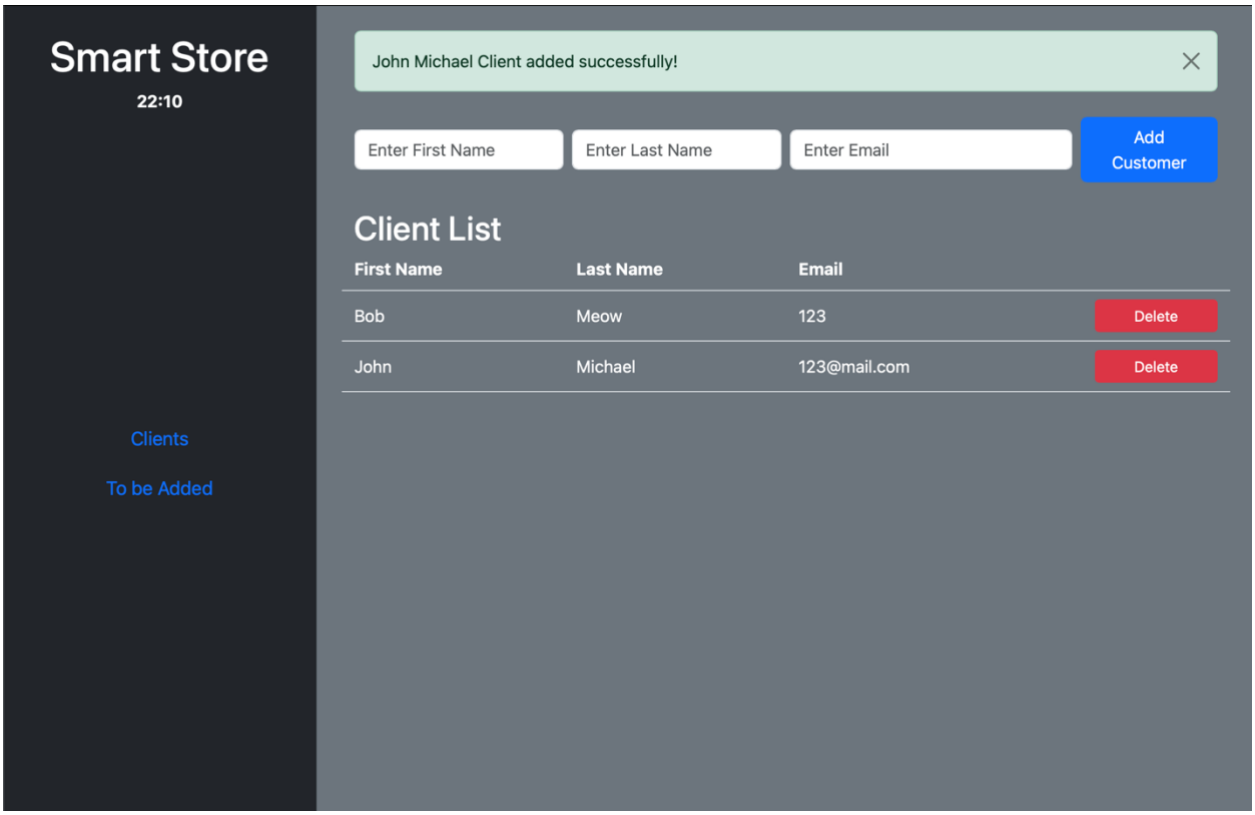
UI Design


Smart Store
12:03
Clients
Something
Logout
Unsuccessful operation
Client Added Successfully
Add
Client List
First Name
Last Name
Email
Ike
Obodoechina
IkeObo96@gmail.com
Smart Store
Clients
Something
Client Added Successfully
Welcome
Insert
Login
Sign Up

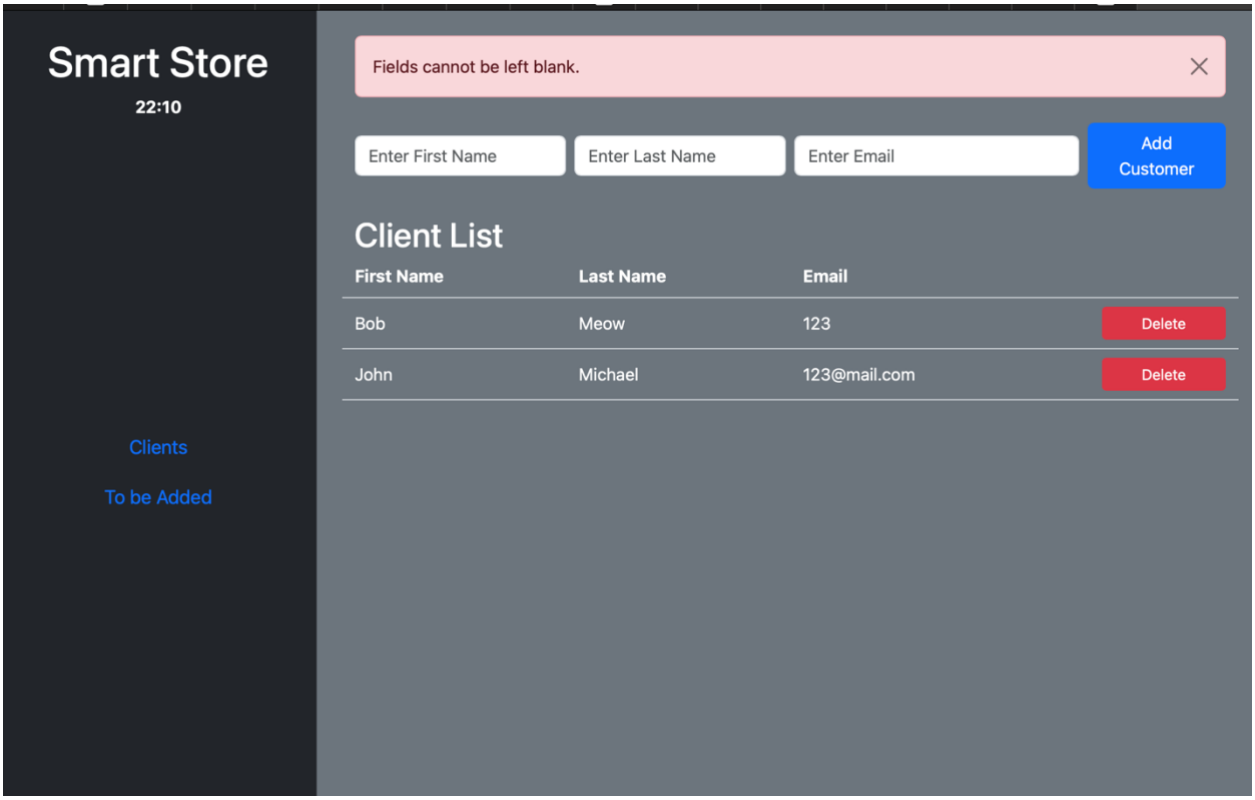
Usage Example (Images)



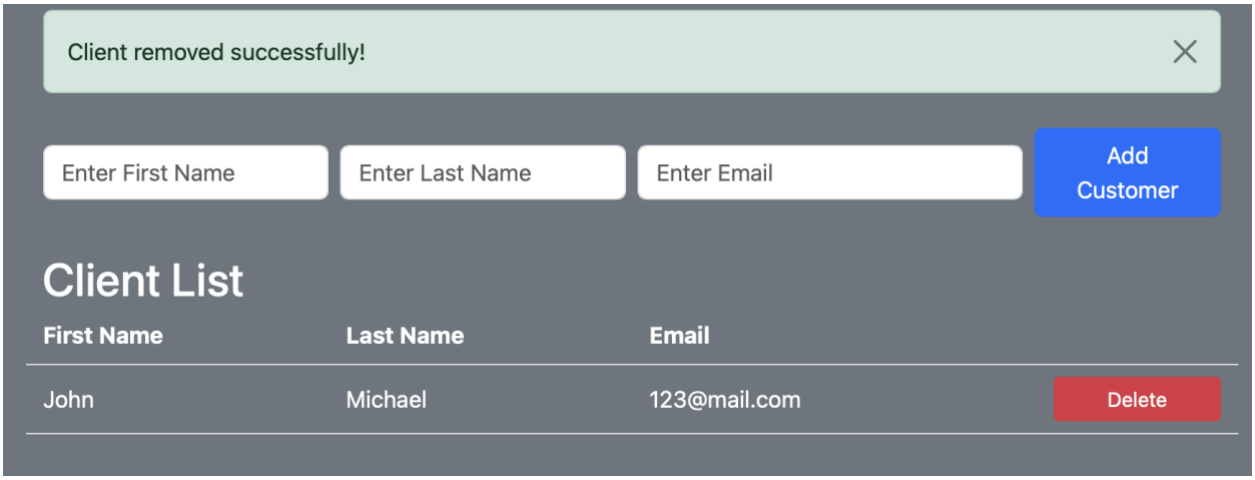
The web page's UI.



Adding a client using the form (successful case)



Adding a client using the form (un-successful case)



Deleting a client